

ETHIO-ITALY POLYTECHNIC COLLEGE



ABC COMPANY NETWORK DESIGN AND IMPLEMENTATION

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Assignment To Tr.

➤ Dershaye Mulugeta

Project Objective

The objective of this project is to design and implement a secure, reliable, and scalable network for ABC Company that connects all computers, printers, and buildings for efficient communication and resource sharing.

Project Scope

- Connecting 180 computers.
- Connecting 9 network printers.
- Interconnecting 3 buildings.
- Designing physical and logical network layouts.
- Configuring routers, switches, and IP addresses.
- Testing and maintaining the network.

Requirement Gathering

There are both Functional and non functional Requirements.

1. Functional Requirements

- Connect all computers.
- Allow communication between all buildings.
- Share printers on each floor.
- Provide Internet access.
- Support file and resource sharing.
- Allow centralized network management.

2. Non-Functional Requirements

The network should:

- Be reliable.
- Be scalable for future expansion.
- Be secure.
- Be easy to maintain.
- Provide high-speed communication.
- Minimize downtime.

Requirement Analysis

ABC Company has three buildings with multiple offices, but no integrated network infrastructure. We would design a Client-Server network using Ethernet LANs within each building and fiber optic links between buildings.

Determine the Network Topology

The selected Topology is Star Topology we selected it because of The advantages :

- Easy to manage.
- Easy to troubleshoot.
- Failure of one computer does not affect others.
- Easy to expand

Determine Network Architecture

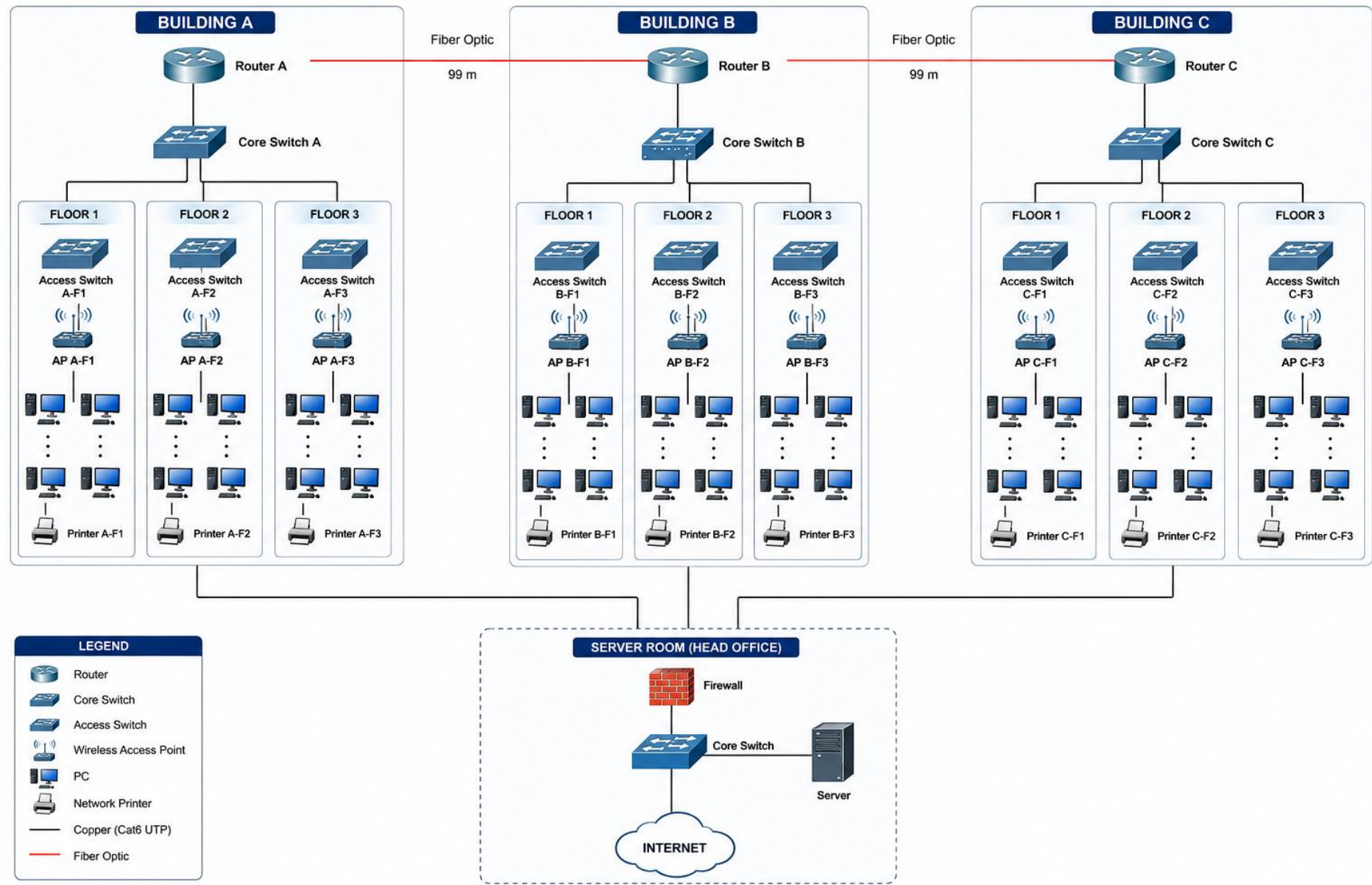
The network architecture that bestly suits this project is Client-server because it offers Centralized management and Easy backup.

It also provides better performance and suitable for Large organisations.

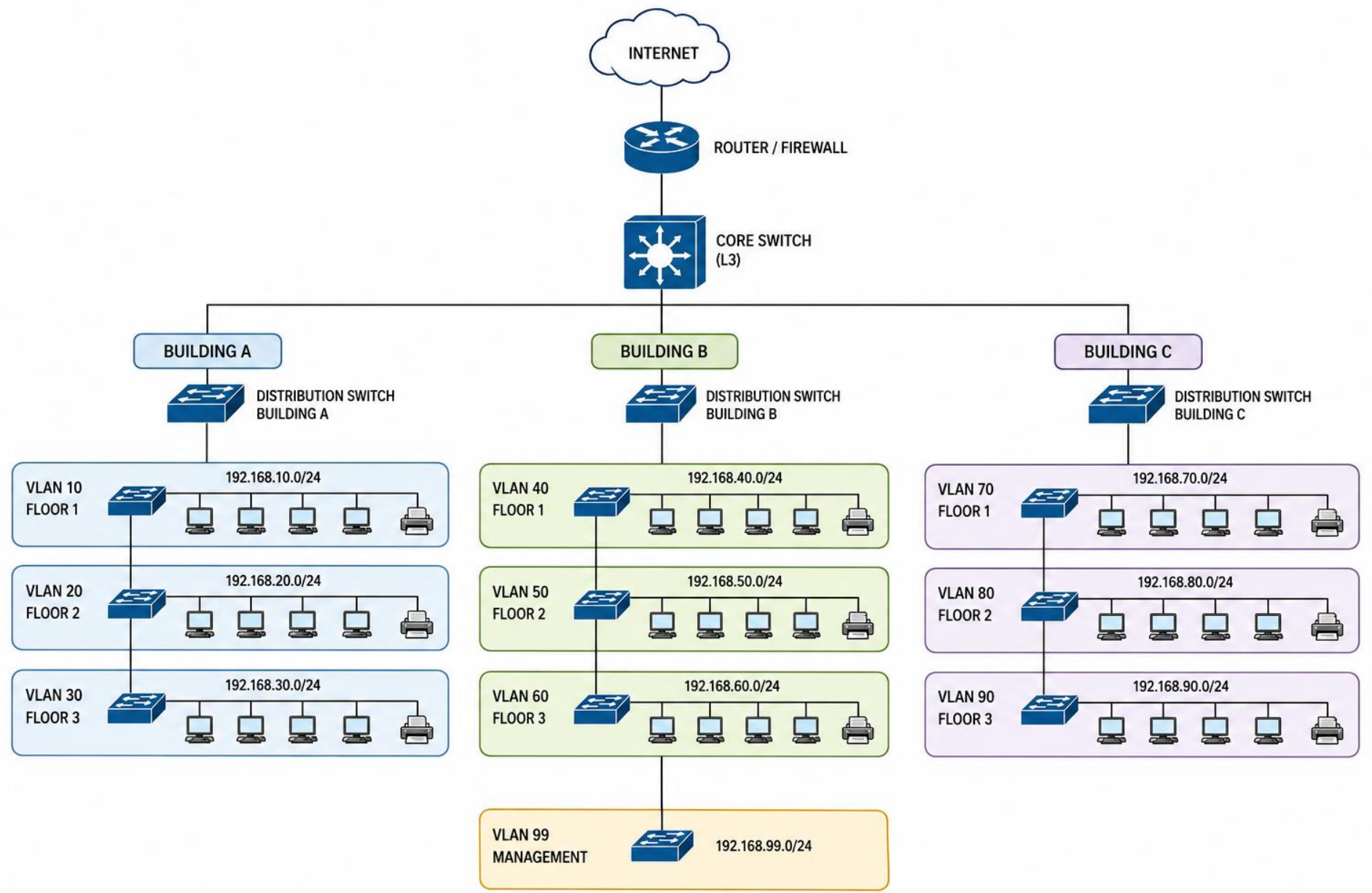
Needed components with their price

No.	Item	Quantity	Unit Price (USD)	Total (USD)	
1	Desktop Computers	180	\$500	\$90,000	
2	Network Printers	9	\$250	\$2,250	
3	Routers (Cisco)	3	\$600	\$1,800	
4	Core Switches (48-Port)	3	\$800	\$2,400	
5	Access Switches (24-Port)	9	\$350	\$3,150	
6	Server	1	\$4,000	\$4,000	
7	Wireless Access Points	9	\$120	\$1,080	
8	Cat6 Cable (Boxes)	20	\$120	\$2,400	
9	Fiber Optic Cable	300 m	\$5/m	\$1,500	
10	RJ45 Connectors	500	\$0.30	\$150	
11	Patch Panels	9	\$100	\$900	
12	UPS	3	\$450	\$1,350	
13	Network Rack Cabinets	3	\$700	\$2,100	
14	Labor (Installation & Configuration)	1	\$8,000	\$8,000	

Physical Diagram of the Network



Logical Diagram of the Network



Time AND Budget

We assume That the project will be finished before The date (Sep, 30, 2026). And based on the plan we would make sure everything finishes on time.

The Estimated Total project cost is ~ USD \$120, 000.

